

ON THE LARGEST PRIME FACTORS OF $p - 1$ AND $p + 1$

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ABSTRACT. We prove that each of the inequalities $P^+(p+1) > P^+(p-1)$ and $P^+(p-1) > P^+(p+1)$ holds for a positive proportion of primes. The argument combines a logarithmic balance identity with a uniform Selberg sieve over a thin cofactor strip. The same transfer mechanism applies to either shift and converts a one-sided large-prime-factor reservoir into an unconditional ordering result in both directions.

1. INTRODUCTION

For $n > 1$, let $P^+(n)$ denote its largest prime factor, and set $P^+(1) = 1$. All logarithms are natural. We write

$$G(x) = \#\{p \leq x : P^+(p+1) > P^+(p-1)\},$$

$$\tilde{G}(x) = \#\{p \leq x : P^+(p-1) > P^+(p+1)\}.$$

where p always denotes a prime.

The comparison of largest prime factors at neighbouring arguments has a long history. The Erdős–Turán conjecture predicts that $P^+(n) < P^+(n+1)$ holds for a set of integers of asymptotic density $1/2$. Erdős and Pomerance [3] established a positive lower density in 1978. Subsequent quantitative refinements include the published bound 0.2017 of Lü and Wang [6]. In a different direction, Teräväinen [9, Theorem 1.16] proved that the logarithmic density is $1/2$, and Tao–Teräväinen [8, Corollary 1.16] obtained the ordinary density $1/2$ at all scales outside an exceptional set of logarithmic density zero. Neither statement asserts the ordinary density limit at every scale. These results concern an unrestricted integer variable; restricting the middle argument to primes introduces an additional distribution problem and does not follow by restriction of an integer-density estimate. For the prime-variable ordering considered here, Wang proved a positive-proportion result under the Elliott–Halberstam conjecture [11]; his survey [10] discusses the prime-variable problem. Our purpose is to obtain unconditional positive-proportion conclusions in both directions.

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Theorem 1.1. *Each ordering occurs for a positive proportion of primes. More precisely,*

$$G(x) \gg \frac{x}{\log x}, \quad \tilde{G}(x) \gg \frac{x}{\log x},$$

and hence

$$\liminf_{x \rightarrow \infty} \frac{G(x)}{\pi(x)} > 0, \quad \liminf_{x \rightarrow \infty} \frac{\tilde{G}(x)}{\pi(x)} > 0.$$

Remark 1.2. The proof permits a numerical positive lower coefficient, but we do not calculate it here since it would be very tiny. For instance, the elementary reservoir estimate in Proposition 2.3 permits $\theta = 11/20$ and $\delta = 1/4$; the remaining sieve and Euler-product constants may be bounded coarsely. Our emphasis is the mechanism rather than a small numerical value. No effective starting point is asserted, since we use the classical Bombieri–Vinogradov theorem.

The key feature is a transfer from one-sided large prime factors to a comparison of two adjacent shifts. The one-sided problem has its own classical development: Goldfeld [4] obtained a positive-proportion result for $P^+(p+a) > x^\theta$, for fixed nonzero a , beyond $\theta = 1/2$, with exponent $0.583\dots$; Baker and Harman [2] later reached 0.677 . The formulation recorded by Akbary–Hambrook [1, Theorem 1.1] supplies, for any fixed $1/2 < \theta \leq 0.677$, a $\delta > 0$ such that

$$\#\{p \leq x : P^+(p+1) > x^\theta\} \geq \delta \frac{x}{\log x}$$

for sufficiently large x . For the proof below, it is enough to obtain the weaker choice $\theta = 11/20$, $\delta = 1/4$ directly, as we do in Proposition 2.3. Such a reservoir alone does not decide which of $P^+(p-1)$ and $P^+(p+1)$ is larger. We bridge that gap by dividing the reverse orderings according to their distance from the diagonal.

Write $Q = P^+(p+1)$ and $R = P^+(p-1)$. Within the reservoir $Q > x^\theta$, a balance identity for the logarithms above $\sqrt{x+1}$ shows that a small number of desired orderings forces a small number of reverse orderings with $R > x^\eta Q$, for any fixed $\eta > 0$. This is an averaged constraint, not a pointwise relation between Q and R . For the remaining reverse orderings, write $p+1 = 2aQ$ and $p-1 = 2bR$. The ratio restriction confines the cofactors to the thin multiplicative strip $b < a \leq 2x^\eta b$. A three-linear-form upper-bound sieve, followed by harmonic summation over this strip, gives

$$\#\{p \leq x : Q(p) > x^\theta, Q(p) < R(p) \leq x^\eta Q(p)\} \ll_\theta \eta \frac{x}{\log x} + \frac{x}{\log^2 x}.$$

The small factor η is the decisive gain: summation over the unrestricted cofactor box would lose it. The same argument works with the roles of $p+1$ and $p-1$ reversed. In that case the reservoir is $R > x^\theta$, the undesired near-diagonal ordering is $R < Q \leq x^\eta R$, and the cofactor relation $aQ - bR = 1$ forces $a < b$; the identical uniform sieve then applies on the reflected thin strip.

A second feature is that the sieve estimate is uniform in cofactors which grow with x . We apply a classical Selberg upper-bound theorem whose constants depend only on fixed local sieve conditions, and verify those conditions and the remainder uniformly. Thus no prime-tuple asymptotic or unverified dependence on growing coefficients is needed. Combining the two controls and choosing a small fixed η completes the transfer.

For comparison, Yang's recent preprint [12, Theorem 1.6] treats $P_y^+(n)$, the largest prime factor of n not exceeding y . It gives a positive-proportion ordering result for shifted primes when $y = x^\alpha$ and $0 < \alpha < 17/32$, which does not directly imply the untruncated ordering considered here. Cofactor sieves also occur in that work; the transfer above relies on their combination with logarithmic balance and a thin strip.

Section 2 establishes the logarithmic estimates and the far-from-diagonal bounds in both directions. Section 3 treats the two reflected near-diagonal sets by the same uniform sieve and thin-strip summation. Section 4 closes both arguments. All external analytic inputs are classical published results.

2. LOGARITHMIC ESTIMATES AND FAR REVERSE ORDERINGS

2.1. The balance identity. For primes $p \leq x$, write

$$Q(p) = P^+(p+1), \quad R(p) = P^+(p-1), \quad Z = \sqrt{x+1},$$

and define

$$L_-(p) = \mathbf{1}_{R(p) > Z} \log R(p), \quad L_+(p) = \mathbf{1}_{Q(p) > Z} \log Q(p).$$

Lemma 2.1. *As $x \rightarrow \infty$,*

$$(1) \quad \sum_{p \leq x} (L_-(p) - L_+(p)) = o(x).$$

In fact the proof gives $O(x \log \log x / \log x)$.

Proof. Let Λ be the von Mangoldt function and let $\pi(x; d, c)$ count primes at most x congruent to c modulo d . Since $\log n = \sum_{d|n} \Lambda(d)$,

$$(2) \quad \sum_{3 \leq p \leq x} \log \frac{p-1}{p+1} = \sum_{d \leq x+1} \Lambda(d) (\pi(x; d, 1) - \pi(x; d, -1)) + \log 3.$$

This is exact for $x \geq 3$: the only correction is removal of $p = 2$, whose contribution is $-\log 3$. There is no discrepancy at $d = x+1$. The left side is $O(\log \log x)$ because $|\log((p-1)/(p+1))| \ll 1/p$.

Apply the classical Bombieri–Vinogradov theorem in its maximal ψ -form, as recorded in [1, Introduction], with saving exponent $A = 3$. Choose its associated fixed B , enlarge it to $B \geq 6$, and put $D = x^{1/2}/(\log x)^B$. Here $\psi(t; d, c) = \sum_{n \leq t, n \equiv c \pmod{d}} \Lambda(n)$. Thus

$$\sum_{d \leq D} \max_{(c,d)=1} \max_{2 \leq t \leq x} \left| \psi(t; d, c) - \frac{t}{\varphi(d)} \right| \ll \frac{x}{\log^3 x}.$$

Removing prime powers costs at most $O(\sqrt{x} \log^2 x)$ per modulus, uniformly for $t \leq x$. Partial summation against $1/\log t$ then gives the π -form below: its kernel has bounded total variation on $[2, x]$, and the summed prime-power and lower-endpoint errors are $O(D\sqrt{x} \log^2 x + D) \ll x/\log^4 x$. Consequently

$$\sum_{d \leq D} \max_{(c,d)=1} \left| \pi(x; d, c) - \frac{\text{Li}(x)}{\varphi(d)} \right| \ll \frac{x}{\log^3 x}.$$

The main terms in the two reduced classes 1, -1 cancel, giving

$$(3) \quad \left| \sum_{d \leq D} \Lambda(d) (\pi(x; d, 1) - \pi(x; d, -1)) \right| \ll \frac{x}{\log^2 x}.$$

For $D < d \leq Z$, Brun–Titchmarsh [1, Lemma 3.1(f), equation (3.11)] gives

$$\pi(x; d, \pm 1) \ll \frac{x}{\varphi(d) \log(x/d)} \ll \frac{x}{\varphi(d) \log x}.$$

The constants are uniform since $d \leq \sqrt{x+1} < x$ and $\log(x/d) \geq (1/2 + o(1)) \log x$. We shall use

$$(4) \quad \sum_{n \leq y} \frac{\Lambda(n)}{\varphi(n)} = \log y + O(1).$$

Indeed, the classical estimate $\sum_{n \leq y} \Lambda(n)/n = \log y + O(1)$ implies this because

$$\sum_q \sum_{k \geq 1} \log q \left(\frac{1}{\varphi(q^k)} - \frac{1}{q^k} \right) = \sum_q \frac{\log q}{(q-1)^2} < \infty.$$

Subtracting (4) at Z and D therefore yields

$$(5) \quad \sum_{D < d \leq Z} \Lambda(d) (\pi(x; d, 1) + \pi(x; d, -1)) \ll \frac{x}{\log x} \left(1 + \log \frac{Z}{D} \right) \ll \frac{x \log \log x}{\log x}.$$

An integer at most $x+1$ has at most one prime divisor greater than Z , and that divisor cannot occur to the second power. Thus the terms $d = q > Z$ in (2) give exactly $L_-(p) - L_+(p)$. The remaining terms $d = q^k > Z$, $k \geq 2$, are bounded in absolute value by a prime-power error E_{pp} . Each class modulo q^k contains at most $x/q^k + 1$ positive integers at most x , so

$$E_{\text{pp}} \ll \sum_{2 \leq k \leq K} \sum_{Z < q^k \leq x+1} \log q \left(\frac{x}{q^k} + 1 \right), \quad K = \left\lfloor \frac{\log(x+1)}{\log 2} \right\rfloor.$$

For $k = 2$, Chebyshev's bound $\sum_{q \leq t} \log q \ll t$ and partial summation give

$$x \sum_{q > (x+1)^{1/4}} \frac{\log q}{q^2} + \sum_{q \leq (x+1)^{1/2}} \log q \ll x^{3/4} + x^{1/2}.$$

For $k \geq 3$, put $y_k = (x+1)^{1/(2k)}$. Uniformly for real $y \geq 1$ and integers $k \geq 2$,

$$(6) \quad \sum_{n>y} \frac{\log n}{n^k} \ll y^{1-k} \log(2y).$$

For completeness, split the integers into $2^j y < n \leq 2^{j+1} y$. The contribution of the j th block is at most an absolute constant times $y^{1-k} 2^{-j(k-1)} (\log(2y) + (j+1) \log 2)$; summing the geometric series proves (6) with a constant independent of k . It follows that the k th contribution is

$$\ll x^{1/2+1/(2k)} \log x + (x+1)^{1/k} \ll x^{2/3} \log x + x^{1/3}.$$

There are $O(\log x)$ such k , whence

$$(7) \quad E_{\text{pp}} \ll x^{3/4} + x^{2/3} \log^2 x = o(x).$$

Combining (2), (3), (5) and (7) proves the lemma. Restoring $p = 2$ costs only $O(1)$. $\square$

2.2. One-sided reservoirs for both shifts.

Lemma 2.2. *For either sign $s \in \{-1, 1\}$, as $x \rightarrow \infty$,*

$$(8) \quad \sum_{p \leq x} \mathbf{1}_{P^+(p+s) > Z} \log P^+(p+s) = \frac{x}{2} + O\left(\frac{x \log \log x}{\log x}\right).$$

For $s = -1$, the term $p = 2$ is understood as zero.

Proof. The proof of Lemma 2.1 also estimates each small-divisor sum separately. For $D = x^{1/2}/\log^B x$, its small-modulus part is

$$\begin{aligned} \sum_{d \leq D} \Lambda(d) \pi(x; d, -s) &= \text{Li}(x) \sum_{d \leq D} \frac{\Lambda(d)}{\varphi(d)} + O(x/\log^2 x) \\ &= \text{Li}(x)(\log D + O(1)) + O(x/\log^2 x) \\ &= \frac{x}{2} + O\left(\frac{x \log \log x}{\log x}\right). \end{aligned}$$

The part $D < d \leq Z$ is bounded by (5). The quantitative prime number theorem in the form $\vartheta(t) = \sum_{p \leq t} \log p = t + O(t/\log t)$ [1, Lemma 3.1(d)] gives $\sum_{3 \leq p \leq x} \log(p+s) = x + O(x/\log x)$: replacing $\log(p+s)$ by $\log p$ costs $O(\log \log x)$. The prime-power discrepancy above Z is bounded by (7). Subtracting the small-divisor sum from the total logarithmic mass therefore gives (8). The $p = 2$ endpoint costs $O(1)$. $\square$

Proposition 2.3. *For $1/2 < \beta < 1$, put*

$$\Delta(\beta) = \frac{1}{2} + 2 \log(2 - 2\beta).$$

Then, for either sign $s \in \{-1, 1\}$,

$$(9) \quad \liminf_{x \rightarrow \infty} \frac{\log x}{x} \#\{p \leq x : P^+(p+s) > x^\beta\} \geq \Delta(\beta).$$

In particular this coefficient is positive for $\beta < 1 - \frac{1}{2}e^{-1/4}$.

Proof. Fix $s \in \{-1, 1\}$ and $\beta > 1/2$. Eventually $x^\beta > Z$. At most one prime factor of $p + s$ exceeds Z , so, apart from the harmless endpoint $p = 2$ when $s = -1$, the logarithmic mass removed between Z and x^β is exactly

$$B_s(x; \beta) = \sum_{Z < q \leq x^\beta} (\log q) \pi(x; q, -s).$$

Brun–Titchmarsh [1, Lemma 3.1(f), equation (3.11)] bounds this uniformly in the sign by

$$2x \sum_{Z < q \leq x^\beta} \frac{\log q}{(q-1) \log(x/q)}.$$

Replacing $q-1$ by q changes this bound by $o_\beta(x)$, since $\log(x/q) \geq (1-\beta) \log x$ and $\sum_{q>Z} \log q/q^2 \ll 1/Z$. Use the Mertens estimate $A(t) = \sum_{q \leq t} \log q/q = \log t + O(1)$. Partial summation against $1/\log(x/t)$ gives

$$\begin{aligned} \sum_{Z < q \leq x^\beta} \frac{\log q}{q \log(x/q)} &= \int_Z^{x^\beta} \frac{dt}{t \log(x/t)} + O_\beta(1/\log x) \\ &= \log \frac{\log(x/Z)}{(1-\beta) \log x} + O_\beta(1/\log x) \\ &= -\log(2-2\beta) + o_\beta(1). \end{aligned}$$

The partial summation error follows because the endpoint values and the total variation of $1/\log(x/t)$ on this interval are $O_\beta(1/\log x)$. Lemma 2.2 now implies, uniformly for the fixed sign s ,

$$\sum_{\substack{p \leq x \\ P^+(p+s) > x^\beta}} \log P^+(p+s) \geq \Delta(\beta)x + o_\beta(x).$$

Each logarithm is at most $\log(x+1)$, proving (9). If $\Delta(\beta) \leq 0$, the resulting lower bound is simply vacuous. $\square$

Remark 2.4. See [1, Theorem 2.5] for the same expression Δ in the quantitative minus-shift setting. The argument above treats both signs directly using standard published inputs: Bombieri–Vinogradov, Brun–Titchmarsh, Mertens’ theorem, and the prime number theorem. An explicit density coefficient does not give an effective starting point, since the BV estimate used in Lemma 2.2 is ineffective.

2.3. Orderings far from the diagonal. Fix $1/2 < \theta < 1$ and define

$$\mathcal{A}_\theta(x) = \{p \leq x : Q(p) > x^\theta\}.$$

For fixed $\eta > 0$, put

$$\mathcal{F}_\eta(x) = \{p \in \mathcal{A}_\theta(x) : R(p) > x^\eta Q(p)\}.$$

Lemma 2.5. *For every fixed $\eta > 0$,*

$$(10) \quad \#\mathcal{F}_\eta(x) \leq \frac{G(x)}{\eta} + o_\eta\left(\frac{x}{\log x}\right).$$

Proof. For sufficiently large x , $x^\theta > Z$. Every prime in $\mathcal{F}_\eta(x)$ contributes to the balance sum an amount

$$\log \frac{R(p)}{Q(p)} > \eta \log x.$$

The truncated logarithm is nondecreasing for $t \geq 1$. Hence primes with $Q(p) \leq R(p)$ contribute nonnegatively, and only primes counted by $G(x)$ can contribute negatively, by at most $\log(x+1)$ each. Therefore

$$\eta \log x \#\mathcal{F}_\eta(x) \leq G(x) \log(x+1) + o(x).$$

After division, replacing $\log(x+1)/\log x$ by 1 costs $O(1/\log x)$, even using the trivial $G(x) \leq x$. This proves (10); the error is for fixed η , as required. $\square$

For the opposite ordering define

$$\tilde{\mathcal{A}}_\theta(x) = \{p \leq x : R(p) > x^\theta\}, \quad \tilde{\mathcal{F}}_\eta(x) = \{p \in \tilde{\mathcal{A}}_\theta(x) : Q(p) > x^\eta R(p)\}.$$

Lemma 2.6. *For every fixed $\eta > 0$,*

$$(11) \quad \#\tilde{\mathcal{F}}_\eta(x) \leq \frac{\tilde{G}(x)}{\eta} + o_\eta\left(\frac{x}{\log x}\right).$$

Proof. Apply the balance identity with the sign reversed. Every prime in $\tilde{\mathcal{F}}_\eta(x)$ contributes to $L_+(p) - L_-(p)$ an amount

$$\log \frac{Q(p)}{R(p)} > \eta \log x.$$

Primes with $R(p) \leq Q(p)$ contribute nonnegatively to this reversed balance sum, while only primes counted by $\tilde{G}(x)$ can contribute negatively, and each such contribution has absolute value at most $\log(x+1)$. Hence

$$\eta \log x \#\tilde{\mathcal{F}}_\eta(x) \leq \tilde{G}(x) \log(x+1) + o(x),$$

which gives (11) exactly as in Lemma 2.5. $\square$

3. A UNIFORM SIEVE OVER A THIN COFACTOR STRIP

3.1. Cofactors and local densities. Put

$$\mathcal{N}_\eta(x) = \{p \in \mathcal{A}_\theta(x) : Q(p) < R(p) \leq x^\eta Q(p)\}.$$

For later use in the opposite direction also put

$$\tilde{\mathcal{N}}_\eta(x) = \{p \in \tilde{\mathcal{A}}_\theta(x) : R(p) < Q(p) \leq x^\eta R(p)\}.$$

For $p \in \mathcal{N}_\eta(x)$, write $q = Q(p)$, $r = R(p)$. Since $q, r > x^\theta > 3$, these factors are odd and we can write uniquely

$$(12) \quad p+1 = 2aq, \quad p-1 = 2br, \quad aq - br = 1.$$

Consequently $(a, b) = 1$, and for $x \geq 1$ the explicit bounds are

$$(13) \quad 1 \leq a, b \leq x^{1-\theta}.$$

Also

$$\frac{a}{b} = \frac{r}{q} \frac{p+1}{p-1}, \quad p+1 \geq q > x^\theta.$$

Thus, for sufficiently large x depending only on θ ,

$$(14) \quad b < a \leq 2x^\eta b.$$

One of $p-1, p+1$ is divisible by 4 and one by 3. Since $q, r > 3$, the remaining factors 2, 3 occur in a or b , so $6 \mid ab$.

Fix a coprime pair satisfying $6 \mid ab$ and (13). The congruences

$$n \equiv -1 \pmod{2a}, \quad n \equiv 1 \pmod{2b}$$

are compatible modulo their common factor 2, and have a unique solution modulo $2ab$. Choose $1 \leq n_0 < 2ab$, and set

$$(15) \quad \begin{aligned} L_1(m) &= bm + B_1, & B_1 &= \frac{n_0 + 1}{2a}, \\ L_2(m) &= 2abm + B_2, & B_2 &= n_0, \\ L_3(m) &= am + B_3, & B_3 &= \frac{n_0 - 1}{2b}. \end{aligned}$$

For triples from (12) these are q, p, r respectively. All intercepts are non-negative integers, and leading coefficients are positive. With $A_1 = b, A_2 = 2ab, A_3 = a$, the determinants are

$$(16) \quad A_1 B_2 - A_2 B_1 = -b, \quad A_1 B_3 - A_3 B_1 = -1, \quad A_2 B_3 - A_3 B_2 = -a.$$

Define

$$(17) \quad H(n) = \prod_{\substack{\ell > 3 \\ \ell \mid n}} \frac{\ell - 1}{\ell - 3}.$$

Lemma 3.1 (Primitive forms and local densities). *The three forms are primitive and pairwise nonproportional. For the number $\rho(\ell)$ of roots of $L_1 L_2 L_3$ modulo ℓ ,*

$$(18) \quad \rho(2) = \rho(3) = 1, \quad \rho(\ell) = \begin{cases} 3, & \ell > 3, \ell \nmid ab, \\ 1, & \ell > 3, \ell \mid ab. \end{cases}$$

In particular $0 < \rho(\ell) < \ell$ at every prime, and

$$(19) \quad \mathfrak{S}(a, b) = \prod_{\ell} (1 - \rho(\ell)/\ell)(1 - 1/\ell)^{-3} = C_3 H(ab),$$

where $C_3 = 9 \prod_{\ell > 3} (1 - 3/\ell)(1 - 1/\ell)^{-3}$ is an absolutely convergent positive constant.

Proof. The defining congruences give $(2ab, n_0) = 1$. If an odd prime divided both b and B_1 , then $n_0 \equiv 1 \pmod{\ell}$ and $n_0 + 1 \equiv 0 \pmod{\ell}$, an impossibility. If $2 \mid b$, a is odd and $n_0 \equiv 1 \pmod{4}$, so $(n_0 + 1)/(2a)$ is odd. Thus $(b, B_1) = 1$. Similarly $(a, B_3) = 1$: when $2 \mid a$, b is odd and $n_0 \equiv -1 \pmod{4}$, so B_3 is odd. The nonzero determinants establish nonproportionality.

If $\ell > 3$ and $\ell \nmid ab$, all three coefficients are invertible and (16) gives three distinct roots. If an odd $\ell \mid a$, then $\ell \nmid b$, $L_2 \equiv -1$ and $L_3 \equiv -b^{-1} \pmod{\ell}$, while L_1 has exactly one root. If an odd $\ell \mid b$, the symmetric calculation gives $L_2 \equiv 1$ and $L_1 \equiv a^{-1} \pmod{\ell}$, with only L_3 having a root. Since exactly one of a, b is divisible by 3, this also proves $\rho(3) = 1$.

If $2 \mid a$, L_2 and L_3 are always odd and L_1 has one root modulo 2. If $2 \mid b$, L_2 and L_1 are always odd and L_3 has one root. This proves (18). The factors at 2, 3 in (19) have product 9; at $\ell > 3$ dividing ab the ratio to the generic factor is $(\ell - 1)/(\ell - 3)$. At the remaining primes the generic factor is $1 + O(\ell^{-2})$, proving convergence. $\square$

3.2. Uniform Selberg sieve. We use Richert's classical upper-bound sieve in the form of [7, Theorem 11.3, p. 135]; see also [5, Chapters 4–5]. The point of using this formulation is that its implied constant depends only on the fixed sieve parameters occurring in (Ω_1) and $(\Omega_2(\kappa))$ (and on the fixed exponent in the condition $z \leq X^A$), not on the coefficients of the linear forms. We record the verification in our family because a and b grow with x .

For real $z > 2$ put $P(z) = \prod_{\ell < z} \ell$ and $F(m) = L_1(m)L_2(m)L_3(m)$. Extend ρ multiplicatively to squarefree integers, with $\rho(1) = 1$. For every squarefree d , the Chinese remainder theorem gives

$$(20) \quad A_d := \#\{1 \leq m \leq M : d \mid F(m)\} = M \frac{\rho(d)}{d} + r_d, \quad |r_d| \leq \rho(d).$$

Indeed, there are exactly $\rho(d)$ relevant residue classes modulo d , and each contains $M/d + O(1)$ integers in $[1, M]$. Thus Richert's remainder condition (R), namely $|r_d| \leq \omega(d)$ for squarefree d , holds with his local-density function $\omega(d) = \rho(d)$.

Lemma 3.2 (Uniform sifted-set bound). *For every fixed $A > 0$, uniformly for $M \geq 2$ and $3 < z \leq M^A$,*

$$(21) \quad \#\{1 \leq m \leq M : (F(m), P(z)) = 1\} \ll_A H(ab) \frac{M}{\log^3 z},$$

where the implied constant is independent of a, b, M, z .

Proof. We apply Richert's Theorem 11.3 with $X = M$ and $\omega(\ell) = \rho(\ell)$. Let us check the hypotheses with constants independent of a and b .

Richert's condition (Ω_1) is

$$0 \leq \frac{\omega(\ell)}{\ell} \leq 1 - \frac{1}{A_1}.$$

By (18),

$$0 \leq \frac{\rho(\ell)}{\ell} \leq \frac{3}{5} = 1 - \frac{1}{5/2},$$

so (Ω_1) holds with the absolute choice $A_1 = 5/2$. His one-sided dimension condition $(\Omega_2(3))$ asks for an absolute A_2 such that, for $2 \leq u \leq v$,

$$\sum_{u \leq \ell < v} \frac{\rho(\ell) \log \ell}{\ell} \leq 3 \log \frac{v}{u} + A_2.$$

Since $\rho(\ell) \leq 3$, this follows immediately from Mertens' estimate

$$\sum_{u \leq \ell < v} \frac{\log \ell}{\ell} = \log \frac{v}{u} + O(1),$$

with an absolute A_2 . Finally, (20) is precisely condition (R).

Richert's theorem therefore gives, for $z \leq M^A$,

$$\#\{1 \leq m \leq M : (F(m), P(z)) = 1\} \ll_A MV(z),$$

where

$$V(z) = \prod_{\ell < z} \left(1 - \frac{\rho(\ell)}{\ell}\right).$$

and the implied constant depends only on A, A_1, A_2 and the fixed sieve dimension 3. In particular, it is uniform in the growing coefficients a, b .

By (18), for $z > 3$,

$$V(z) = \frac{1}{3} \prod_{3 < \ell < z} (1 - 3/\ell) \prod_{\substack{3 < \ell < z \\ \ell \nmid ab}} \frac{\ell - 1}{\ell - 3} \ll \frac{H(ab)}{\log^3 z}.$$

Here Mertens' product formula, together with the absolutely convergent generic Euler product in (19), gives the first factor $\ll (\log z)^{-3}$; extending the second product to all prime divisors of ab only increases it. This proves (21). $\square$

Lemma 3.3. *Fix $1/2 < \theta < 1$. Uniformly for coprime a, b satisfying $6 \mid ab$ and (13), let $N_\theta(a, b; x)$ count integers $0 \leq m \leq M$, where*

$$(22) \quad M = \left\lfloor \frac{x - n_0}{2ab} \right\rfloor,$$

for which all three values in (15) are prime and $L_1(m), L_3(m) > x^\theta$. Then

$$(23) \quad N_\theta(a, b; x) \ll_\theta H(ab) \frac{x}{ab \log^3 x}.$$

Proof. Since $n_0 < 2ab$ and $ab \leq x^{2-2\theta}$,

$$(24) \quad M = \frac{x}{2ab} + O(1) \asymp \frac{x}{ab}, \quad M \gg x^{2\theta-1}, \quad \log M \asymp_\theta \log x.$$

All constants here are uniform in a, b once x is sufficiently large in terms of the fixed θ . In particular $M \rightarrow \infty$ uniformly. The single endpoint $m = 0$ costs at most 1.

Put

$$\kappa = \frac{2\theta - 1}{8}, \quad z = x^\kappa.$$

Then $0 < \kappa < \theta$. Moreover (24) gives, uniformly in a, b ,

$$z^4 = x^{(2\theta-1)/2} \leq M$$

for all sufficiently large x depending only on θ ; hence $z \leq M^{1/4}$. Thus Lemma 3.2 applies with the fixed choice $A = 1/4$.

For a triple counted by $N_\theta(a, b; x)$ with $m \geq 1$, $L_1, L_3 > x^\theta$, while from (15)

$$L_2 = 2aL_1 - 1 > x^\theta - 1 > z$$

for sufficiently large x . Since all three values are prime and exceed z , their product is coprime to $P(z)$. Lemma 3.2 therefore gives

$$N_\theta(a, b; x) \ll_\theta 1 + H(ab) \frac{M}{\log^3 z} \ll_\theta 1 + H(ab) \frac{x}{ab \log^3 x}.$$

Finally, $H(ab) \geq 1$ and (24) imply $x/(ab \log^3 x) \rightarrow \infty$ uniformly, so the endpoint term is absorbed. This proves (23). $\square$

3.3. Harmonic summation over the strip.

Lemma 3.4 (Mean and interval harmonic bounds). *There is an absolute constant C_H such that, for $T \geq 1$,*

$$(25) \quad A_H(T) := \sum_{n \leq T} H(n) \leq C_H T, \quad \sum_{n \leq T} \frac{H(n)}{n} \leq C_H (1 + \log T).$$

For all real $1 \leq U < V$,

$$(26) \quad \sum_{U < n \leq V} \frac{H(n)}{n} \leq C_H \left(1 + \log \frac{V}{U} \right).$$

Proof. Let c be multiplicative, with $c(1) = 1$, $c(2) = c(3) = 0$, $c(\ell) = 2/(\ell-3)$ for primes $\ell > 3$, and $c(\ell^k) = 0$ for every prime ℓ and $k \geq 2$. Expanding the local factors gives $H(n) = \sum_{d|n} c(d)$. The coefficients are nonnegative, so

$$A_H(T) = \sum_{d \leq T} c(d) \left\lfloor \frac{T}{d} \right\rfloor \leq T \sum_{d \geq 1} \frac{c(d)}{d} = C_H T,$$

where

$$C_H = \prod_{\ell > 3} \left(1 + \frac{2}{\ell(\ell-3)} \right) < \infty.$$

For real endpoints partial summation is exactly

$$\sum_{U < n \leq V} \frac{H(n)}{n} = \frac{A_H(V)}{V} - \frac{A_H(U)}{U} + \int_U^V \frac{A_H(t)}{t^2} dt.$$

Discard the nonpositive term and use $A_H(t) \leq C_H t$ to obtain (26). For the full harmonic sum, add $H(1) = 1 = A_H(1)$ to the exact identity with $U = 1$.

The negative boundary term cancels, leaving $A_H(T)/T + \int_1^T A_H(t)t^{-2} dt \leq C_H(1 + \log T)$. This includes $T = 1$ and does not enlarge C_H . $\square$

Proposition 3.5. *For fixed $1/2 < \theta < 1$ and every $\eta > 0$,*

$$(27) \quad \#\mathcal{N}_\eta(x) \ll_\theta \eta \frac{x}{\log x} + \frac{x}{\log^2 x}.$$

The implied constant and a sufficiently large threshold for x are independent of η .

Proof. Each $p \in \mathcal{N}_\eta(x)$ determines a unique (a, b) satisfying (13), (14), $(a, b) = 1$, $6 \mid ab$. Lemma 3.3, applied before discarding the restrictions, gives

$$\#\mathcal{N}_\eta(x) \ll_\theta \frac{x}{\log^3 x} \sum_{\substack{b < a \leq 2x^\eta b \\ a, b \leq x^{1-\theta} \\ (a, b) = 1, 6 \mid ab}} \frac{H(ab)}{ab}.$$

For every positive integer pair, $H(ab) \leq H(a)H(b)$; shared prime factors can only increase the right side. With $T = x^{1-\theta}$, we can now discard the arithmetic restrictions and the upper bound on the inner variable a . Lemma 3.4 gives

$$\begin{aligned} \sum_{\substack{b < a \leq 2x^\eta b \\ a, b \leq T}} \frac{H(ab)}{ab} &\leq \sum_{b \leq T} \frac{H(b)}{b} \sum_{b < a \leq 2x^\eta b} \frac{H(a)}{a} \\ &\leq C_H(1 + \log 2 + \eta \log x) \sum_{b \leq T} \frac{H(b)}{b} \\ &\ll \log x(1 + \eta \log x). \end{aligned}$$

All constants here are absolute (apart from the fixed θ in the sieve). Multiplication by $x/\log^3 x$ proves (27). $\square$

Proposition 3.6 (Reflected thin-strip bound). *For fixed $1/2 < \theta < 1$ and every $\eta > 0$,*

$$(28) \quad \#\tilde{\mathcal{N}}_\eta(x) \ll_\theta \eta \frac{x}{\log x} + \frac{x}{\log^2 x}.$$

The implied constant and a sufficiently large threshold for x are independent of η .

Proof. Let $p \in \tilde{\mathcal{N}}_\eta(x)$ and write $q = Q(p)$, $r = R(p)$ and

$$p + 1 = 2aq, \quad p - 1 = 2br, \quad aq - br = 1.$$

Here $q > r > x^\theta$, so the box bounds (13), coprimality $(a, b) = 1$, and $6 \mid ab$ are unchanged. Moreover $a < b$. Indeed, if $a \geq b$, then, since q and r are distinct odd primes,

$$1 = aq - br = a(q - r) + (a - b)r \geq q - r \geq 2,$$

a contradiction. Also

$$\frac{b}{a} = \frac{q}{r} \frac{p-1}{p+1} < \frac{q}{r} \leq x^\eta,$$

so the cofactors lie in the reflected strip

$$(29) \quad a < b \leq x^\eta a.$$

For each such pair the same linear forms (15) have prime values q, p, r , and Lemma 3.3 applies without any change. Consequently, after discarding $(a, b) = 1$ and $6 \mid ab$ and using $H(ab) \leq H(a)H(b)$, with $T = x^{1-\theta}$ we have

$$\#\tilde{\mathcal{N}}_\eta(x) \ll_\theta \frac{x}{\log^3 x} \sum_{\substack{a < b \leq x^\eta a \\ a, b \leq T}} \frac{H(a)H(b)}{ab}.$$

Using Lemma 3.4 first in the b -variable gives

$$\begin{aligned} \sum_{\substack{a < b \leq x^\eta a \\ a, b \leq T}} \frac{H(a)H(b)}{ab} &\leq \sum_{a \leq T} \frac{H(a)}{a} \sum_{a < b \leq x^\eta a} \frac{H(b)}{b} \\ &\leq C_H(1 + \eta \log x) \sum_{a \leq T} \frac{H(a)}{a} \\ &\ll \log x(1 + \eta \log x). \end{aligned}$$

Multiplication by $x/\log^3 x$ proves (28). $\square$

4. COMPLETION OF THE ARGUMENT

Proposition 4.1 (Square-root-breaking criterion). *Fix $1/2 < \theta < 1$. Suppose there is $\delta > 0$ such that, for all sufficiently large x ,*

$$(30) \quad \#\mathcal{A}_\theta(x) \geq \delta \frac{x}{\log x}.$$

Then $G(x) \gg_{\theta, \delta} x/\log x$.

Proof. For $p > 3$, equality $Q(p) = R(p)$ is impossible. A common largest prime factor would divide 2, making both $p-1, p+1$ powers of 2. If $p-1 = 2^u$ and $p+1 = 2^v$, then $2^u(2^{v-u} - 1) = 2$, forcing $u = 1, v = 2, p = 3$. Thus, up to a bounded endpoint exception,

$$\mathcal{A}_\theta(x) \subseteq \{p \leq x : Q(p) > R(p)\} \cup \mathcal{F}_\eta(x) \cup \mathcal{N}_\eta(x).$$

By (30), Lemma 2.5 and Proposition 3.5, there is $C_\theta > 0$, independent of η , such that for every fixed $\eta > 0$,

$$(31) \quad \delta \frac{x}{\log x} \leq \left(1 + \frac{1}{\eta}\right) G(x) + C_\theta \eta \frac{x}{\log x} + o_{\theta, \eta} \left(\frac{x}{\log x}\right).$$

Fix $\eta > 0$ with $C_\theta \eta < \delta/2$. Then take x sufficiently large that the last error has absolute value at most $\delta x/(4 \log x)$. It follows explicitly that

$$G(x) \geq \frac{\delta \eta}{4(1 + \eta)} \frac{x}{\log x}.$$

The order of choices is θ, δ , then C_θ , then a fixed η , and finally $x \rightarrow \infty$; no error estimate uniform as $\eta \rightarrow 0$ is used. $\square$

Proposition 4.2 (Reverse square-root-breaking criterion). *Fix $1/2 < \theta < 1$. Suppose there is $\delta > 0$ such that, for all sufficiently large x ,*

$$(32) \quad \#\tilde{\mathcal{A}}_\theta(x) \geq \delta \frac{x}{\log x}.$$

Then

$$\tilde{G}(x) \gg_{\theta, \delta} \frac{x}{\log x}.$$

Proof. For $p > 3$ equality $Q(p) = R(p)$ is impossible, as observed in the proof of Proposition 4.1. Hence, up to a bounded endpoint exception,

$$\tilde{\mathcal{A}}_\theta(x) \subseteq \{p \leq x : R(p) > Q(p)\} \cup \tilde{\mathcal{F}}_\eta(x) \cup \tilde{\mathcal{N}}_\eta(x).$$

Using Lemma 2.6 and Proposition 3.6, we obtain, for every fixed $\eta > 0$,

$$\delta \frac{x}{\log x} \leq \left(1 + \frac{1}{\eta}\right) \tilde{G}(x) + C'_\theta \eta \frac{x}{\log x} + o_{\theta, \eta} \left(\frac{x}{\log x}\right),$$

where C'_θ is independent of η . Choosing η and then x in the same order as in Proposition 4.1 proves the claim. $\square$

Proof of Theorem 1.1. Use Proposition 2.3 with $\theta = 11/20$. The bound $-\log(9/10) \leq 1/9$ gives $\Delta(11/20) > 1/4$. For $s = 1$ the strict liminf margin yields (30) with $\delta = 1/4$ for all sufficiently large x , so Proposition 4.1 gives $G(x) \gg x/\log x$.

For $s = -1$ the same proposition gives (32), again with $\delta = 1/4$ for all sufficiently large x . Proposition 4.2 therefore gives $\tilde{G}(x) \gg x/\log x$. The prime number theorem converts both estimates into the asserted positive lower proportions. $\square$

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